

Ring / Johnson

A shift register with feedback becomes a counter without a binary adder

Ring walk starter

- Starter: ring mode with one-hot pattern zero-zero-zero-one
- Step once and the one moves to zero-zero-one-zero
- Four steps complete a full ring period back to zero-zero-zero-one
- Switch to Johnson: load zero-zero-zero-zero, step eight times for the full two-N cycle
- Watch the feedback box, plain q-three for ring, not-q-three for Johnson
- Beware all-zero on a ring, it can lock with zero feedback forever

Browser lab

Digital Design and Verification Platform

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INTERACTIVE TOOL

Ring / Johnson counter

Step a 4-bit **ring** (circulating one-hot) and **Johnson** (twisted-ring) counter — same shift, different feedback.

Starter example: ring counter with one-hot 0001 — step to walk the 1; switch to Johnson for the twisted sequence.

Load starter example

Challenges (1/22)

Quiz: ring. Circulating one-hot counter is a? Answer: ring

Answer

Show hint

Check

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Idle

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quiz-period-j

starter

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full-ring

johnson-step

full-johnson

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stuck-zero

one-hot-check

demo

explain

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code-ring

code-johnson

fb-invert

full-path

Core ideas

Ring

Shift + feed MSB back — circulating one-hot (N states).

Johnson

Shift + feed *inverted* MSB — twisted ring (2N states).

Ring Johnson starter

Workbook practice

- In the workbook track, write the always block for ring versus Johnson feedback
- List four ring states starting from zero-zero-zero-one
- Count Johnson states for four bits, why two-N?
- Sketch what happens if a ring starts at all-zero
- Name one pitfall: invalid init or illegal states without recovery

Pitfalls to watch

- Do not confuse display order, here q-three through q-zero is MSB left
- Ring needs valid one-hot init; Johnson needs the right reset
- Neither replaces a binary counter when you need full two-to-the-N counting
- And remember: the browser lab is literacy
- Real designs still need timing, enable, and illegal-state handling

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, finish a few challenges after the starter
- On paper, draw one ring walk and one Johnson feedback inversion
- When you are ready, take the short quiz, then continue to LFSR and PRBS